CO05	**Assess** and justify the use of data visualization techniques and R programming tools for effective data analysis, interpretation, and communication of actionable insights.	L5	60

• 8. CO: PO Mapping (3 – Strong, 2 – Medium, 1 – Low Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	1	-	-	-	1	-	
CO2	3	3	3	3	1	1	-	
CO3	3	3	3	3	-	-	-	
CO4	2	3	2	3	-	2	-	1
CO5	2	-	-	2	2	2	2	1
Articulation	2.6	2.5	2.6	2.75	1.5	1.5	1	1

• 9. Syllabus

Module	Topics	Contact Hours
I	INTRODUCTION TO BIG DATA: Introduction to Big Data - Dimensions of Big Data, Challenges of Conventional Systems, Features of Big Data - Security, Compliance, Auditing and Protection, Big Data Characteristics - Analytic Processes and Tools, Analysis vs Reporting, Modern Data Analytic Tools, Statistical Concepts: Sampling Distributions, ReSampling - Statistical Inference - Prediction Error.	9
II	DATA ANALYSIS: Regression Modelling – Multivariate, Support Vector and Kernel Methods; Analysis of Time Series: Linear Systems Analysis, Correlation & Reliability, Regression – Analysis of Variance (ANOVA), Decision Trees – Association, Classification Trees.	9
III	MINING DATA STREAMS: The Stream Data Model - Sampling of Streams, Filtering Streams, Bloom Filter, Counting Distinct Elements - Estimating the number of 1's in a window, Data Stream Mining, Open Source Tools.	9
IV	PREDICTIVE ANALYSIS: Predictive Analytics – Supervised and Unsupervised Learning, Mining Frequent Itemsets - Market Based Model, Apriori Algorithm – Handling large data sets in Main memory, Limited Pass algorithm - Counting frequent itemsets in a stream, Clustering Techniques – Hierarchical, K-Means – Clustering High Dimensional Data.	9
V	VISUALIZATION AND R-PROGRAMMING: Visualization – Purpose of Visualization, Tree and Graph and Time series data visualization techniques, Basic Interaction Techniques. R Fundamentals - Installing, Loading and Running R programs, Accessing Data and R-Script.	9